

In information retrieval, Okapi BM25 is a ranking function used by search engines to estimate the relevance of documents to a given search query. It is based on the probabilistic retrieval framework developed in the 1970s and 1980s by Stephen Robertson, Karen Sparck Jones, and others. The name of the actual ranking function is BM25, where BM stands for best matching. The Okapi prefix comes from the Okapi information retrieval system implemented at London's City University in the 1980s and 1990s, in which the function was first used.

BM25 and its newer variants represent state-of-the-art retrieval functions used in document retrieval. It is a bag-of-words retrieval function that ranks a set of documents based on the query terms appearing in each document, regardless of their proximity within the document. It is a family of scoring functions with slightly different components and parameters.

The function is best understood as an extension of the binary independence model that adds two things the earlier model lacked: term frequency and document length normalization. The binary independence model was designed for short catalogue records of fairly consistent length, where those two factors matter little. For modern full-text collections they matter a great deal.

Term frequency enters through a saturating function controlled by the parameter k_1 . As the frequency of a term in a document grows, its contribution to the score increases, but with diminishing returns, approaching an asymptote rather than growing without bound. The parameter b controls document length normalization: at b equal to one, term frequency is fully normalized by the ratio of document length to average document length, and at b equal to zero no normalization is applied. Typical empirical values are k_1 between 1.2 and 2.0 and b around 0.75.

A third parameter, k_3 , applies the same saturation to the query side and matters only for long queries, where a term may be repeated. For the short queries typical of interactive search it has almost no effect on the ranking.